

Program Design Document

Program Name

Calculator Program

Purpose

This program allows the user to input two integers and choose a mathematical operation. It then calculates and displays the result. The program solves the problem of performing basic arithmetic operations in a simple and interactive way while handling errors such as division by zero.

Inputs

- First integer (c)
 - Second integer (d)
 - Operator (op): one of the following characters: +, -, *, /, %
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Outputs

- Result of the calculation
 - OR an error message if:
 - The operator is invalid
 - Division or modulus by zero is attempted
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Variables

Variable	Type	Purpose
c	int	Stores first number
d	int	Stores second number
op	char	Stores operator
result	int	Stores result of calculation

Key Design Choices

- I used an **if-else statement** to determine which operation to perform because each operator represents a different condition and only one should be executed.

- I used a **char variable for the operator** because mathematical symbols like + and - are characters, not integers.
 - I included **error checking for division by zero** to prevent runtime errors and ensure safe execution.
 - I did not use loops because the program only needs to perform one calculation per run.
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Program Steps (Algorithm)

1. Prompt the user to enter the first number (c).
 2. Prompt the user to enter the second number (d).
 3. Prompt the user to enter an operator (+, -, *, /, %).
 4. Check the operator:
 - If op is '+', compute $c + d$
 - Else if op is '-', compute $c - d$
 - Else if op is '*', compute $c * d$
 - Else if op is '%':
 - If $d == 0 \rightarrow$ display error
 - Else compute $c \% d$
 - Else if op is '/':
 - If $d == 0 \rightarrow$ display error
 - Else compute c / d
 - Else:
 - Display "Invalid operator"
 5. Display the result
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Functions

Function: main()

Purpose: Handles user input, determines the operation, performs the calculation, and displays the result.

Sample Input/Output

Input:

First number: 10

Second number: 5

Operator: +

Output:

Result: 15

Testing

Test Case 1

Input: 10, 5, +

Expected Result: 15

Test Case 2

Input: 10, 0, /

Expected Result: Error message "Cannot divide by 0"

Test Case 3

Input: 7, 3, *

Expected Result: 21

Test Case 4

Input: 8, 2, %

Expected Result: 0

Test Case 5

Input: 5, 5, x

Expected Result: "Invalid operator"